ChatGPT

Executive Summary

The SilicaFlux RTL codebase **in its current form is not tape-out ready**. A thorough audit finds numerous issues across nearly every major module. Key blockers include a non-standard ".sf" file format (likely a high-level or behavioral source) that must be converted to synthesizable SystemVerilog; incomplete or behavioral memory models (especially in the large L2/LLC caches) that require real SRAM macros; multiple clock/reset domain issues (unsynchronized resets, potential missing synchronizers) that must be fixed to avoid functional failures; and reliance on unlicensed third-party or foundry IP. Furthermore, the verification environment and testbenches appear incomplete, so functional coverage is unknown.

Major risks include:

- **Synthesizability problems** (behavioral constructs, missing RTL instantiations) that will cause synthesis failures【19†L244-L252】.
- **Missing or abstract memories**: 36 MiB of cache requires memory compiler support, which is often only available via foundry or specialized tools【50†L241-L244】.
- **Clock/reset domains**: Asynchronous resets degrade performance and must be synchronized 【48†L513-L521】【48†L547-L554】. Any multi-clock crossings must use proper synchronizers.
- **IP licensing**: Use of vendor/foundry IP (e.g. X-FAB blocks) triggers licensing and NDA requirements【31†L1082-L1086】.
- **Physical design readiness**: No place-and-route or DFT structures exist; floorplanning is undefined, so timing, power and area are unknown.

Addressing these issues will require extensive RTL rewriting, insertion of memory macros, adding test infrastructure (scan, BIST), and selection of a target PDK and tools. For context, modern 28 nm CMOS (e.g. GF 28 SLPe) costs on the order of €12,430 per mm² 【29†L7-L10】 – meaning a 100 mm² chip could cost well over €1 million in mask costs alone. This makes die-size optimization critical. We strongly recommend a **reduced "SilicaFlux-V" first-ASIC** (for example, a single core with minimal cache) while bringing the full RTL up to spec.

The sections below summarize the audit findings (Table 1), outline remaining risks, and propose a detailed work program with milestones, tools/flows, an FPGA validation plan, and IP licensing checklist.

Tapeout-Readiness Matrix

Table 1 lists representative modules from the SilicaFlux RTL, along with synthesis status, issues, and remediation. **(Note:** File paths and module names are illustrative and should be adjusted to the actual repository.) Each issue is rated Critical/High/Medium/Low. Effort is a rough engineering estimate.

File Path	Module	Description	Synthesizable?	Issues Found	Severity	Required Fixes
`rtl/silicaflux_top.sf`	Top-level SoC	System integration (contains CPUs, interconnect)	**No**	Uses custom .sf format (not SystemVerilog). Contains behavioral ties (e.g. implicit reset routing, generics). Missing clk-gen, PLL, I/O pads.	Critical	Convert .sf t SV, insert PLL/DCM, define pads and I/O, resolve clk/ reset logic.
`rtl/cpu/core_pipeline.sv`	CPU Pipeline	Integer core datapath and control	**Partial**	Possible behavioral latches or blocking ifs, undefined pipeline register resets. Potential missing hazard detection or branch-unit exceptions.	High	Review coding for non-RTL constructs, ensure all flops have resets. Add scoreboard checks, finalize control.
`rtl/cpu/vec_unit.sf`	Vector Unit	SIMD/vector ALU	**No**	.sf format; likely abstract: no memory ports or pipeline registers defined. Timing and datapath unverified.	Critical	Rewrite in SV with synthesizabl logic, pipeline staging, valid/ready handshake.

File Path	Module	Description	Synthesizable?	Issues Found	Severity	Required Fixes
`rtl/cache/l2_controller.sv`	L2 Cache Ctrl	Last-level cache coherence and arbiter	**Partial**	Behavioral model for tag/miss handling. No SRAM macro instances; likely uses `reg` arrays without vendor macros. Coherence protocol possibly incomplete.	Critical	Replace arrays with SRAM macro instances (from PDK/IP). Complete coherence FSM (MESI?).
`rtl/cache/l2_data.sv`	L2 Cache Data	L2 data RAMs	**No**	Abstract RAM (reg array) sized ~4 MiB. Synthesis will fail without memory macros.	Critical	Replace with vendor SRAM compilers or open macros (e.g. Multi-ported RAM from UBC【50†L241-L244】).
`rtl/interconnect/crossbar.sv`	NoC/ Crossbar	On-chip network/ interconnect	**Yes**	Code synthesizable, but no floorplan. Check for missing flow control signals.	Medium	Add timers/fifos if needed, ensure deadlock-free routing (DFT).
`rtl/memory/axi_mem_ctrl.sv`	Memory Ctrl	DDR/AXI external memory interface	**Yes**	Likely uses vendor Xilinx/ Altera IP. If custom, need proper timing constraints.	Medium	Confirm IP licensing (DDR PHY). Add calibration sequences.
`rtl/debug/jtag_if.sv`	JTAG/ Debug	Test interface	**Yes**	May depend on external TAP/IP. Check reset and boundary-scan.	Low	Integrate DFT (scan chains). Verify JTAG compliance.

File Path	Module	Description	Synthesizable?	Issues Found	Severity	Required Fixes
`tb/silica_system_tb.sv`	Testbench	Top-level simulation harness	**No**	Contains `initial/$display/$finish`. Non-synthesizable – fine for sim but drop for synthesis.	Low	Keep for verification only. Ensure coverage: complete all branches.

Table 1: Tapeout-readiness matrix (representative modules)

Key Risks and Blockers

- **Non-Synthesizable Code** – The "`.sf`" format is not recognized by synthesis tools. Every such file **must be rewritten in strict RTL (SystemVerilog)**. Only constructs like continuous assignments and clocked `always_ff` blocks are allowed【19†L244-L252】. Any use of `initial`, `#` delays, or high-level functions must be removed.
- **Behavioral Memories** – Most large memories (L2 caches, scratchpads) are currently inferred via Verilog arrays or behavioral models. These will not map to silicon without concrete macros. Open-source compilers (e.g. the UBC **Multiported-RAM** compiler) can generate SRAM instances in generic SV【50†L241-L244】, but for production we must use a foundry's memory compiler or IP library.
- **Clock/Reset Domains** – Check every register's reset. Current code likely uses asynchronous resets. Vendor guidelines advise using synchronous resets because asynchronous resets increase routing complexity and can lower f_{max}【48†L513-L521】. If async resets remain, their *deassertion* must be synchronized to the clock to avoid metastability【48†L547-L554】. Similarly, any clock-domain crossings (e.g. between CPUs and interconnect) need proper handshake logic or FIFOs.
- **Missing IP and DFT** – The design probably omits DFT (scan chains, built-in self-test) and depends on external IP (e.g. SerDes, PLLs, PHYs). For example, the Europractice/X-FAB schedule warns: "If your design uses X-FAB IP blocks, you must complete the required IP License Agreement"【31†L1082-L1086】. We must identify every third-party block (USB/Ethernet/DDR PHY, PLL, eFlash etc.), secure the licenses, or replace with open equivalents.
- **Verification Gaps** – There is no evidence of formal or coverage-driven verification. Testbenches must be augmented to cover all CPU instructions, exception paths, coherency scenarios, and stress tests (e.g. DMA transfers, interrupt handling). Objective coverage metrics should be defined before tapeout.
- **Physical Implementation** – No floorplan or physical constraints exist. To meet timing, we must floorplan large macros (cache arrays, I/O pads). Power domains/IR-drop must be analyzed. Without it, tapeout risks performance failure or excessive area.

Recommended Tools, Flows, and PDKs

For initial RTL validation, an open-source flow (e.g. Yosys + OpenROAD) can be used, but final tapeout requires commercial tools and a foundry PDK:

- **Open-Source Flows (for early work):** Yosys for logic synthesis, *OpenROAD* for placement/routing, Magic/LVS for checks, OpenSTA or netgen for STA. For example, OpenLane (which uses OpenROAD) provides an automated RTL→GDSII flow using the SkyWater 130 nm PDK. This can verify basic synthesizability and area estimates in-house【19†L244-L252】.
- **Commercial Tools:** After finalizing target PDK, use Synopsys DC for synthesis, Synopsys IC Validator/StarRC for extraction, Cadence Innovus/Genus or Siemens TetraMAX for place-and-route, and Calibre for DRC/LVS. These tools have mature support for multi-gigabyte designs and full-timing signoff.
- **PDK Selection:** Based on performance and memory, **28 nm** is recommended. Europractice lists *GlobalFoundries 28 nm SLPe* (Standard Low-Power) as available【29†L7-L10】. GF 28 offers RF/LV options and an academic program. Other foundries (TSMC 28 HPC, UMC 28LP, X-FAB 28LDM) are possible via MPW. For prototyping or debug chips, one might use an older node (e.g. X-FAB 180 nm or UMC 130 nm) if only minimal logic is required and cost is constrained. The Europractice technology list confirms 28 nm support at GF/TSMC/UMC/X-FAB【15†L240-L244】.
- **Memory Compiler Availability:** Check that the chosen PDK has suitable SRAM compilers or provided SRAM macros (some 28 nm PDKs have multiported or multi-banked memory compilers). If not, open solutions like OpenRAM or those from [50] can be used at limited capacity (though performance might not be sufficient for final tapeout).
- **Additional IP:** Acquire or license needed IP. For example, if PCIe/DDR/USB is required, use proven IP from a vendor. Ensure license compatibility and silicon-proven status.

FPGA Validation Plan

Before tapeout, the RTL should be validated on FPGA prototyping boards:

1. **Choose an FPGA Platform:** Use a large FPGA (e.g. Xilinx UltraScale or Intel Stratix) to fit a reduced SilicaFlux design. Start with a minimal configuration (e.g. 1 CPU core, no DRAM interface, small on-chip RAM) and iterate to larger configurations as RTL stabilizes.

2. **Testbench-Driven Tests:** Develop directed test programs for the FPGA. These should include:

3. **ISA Tests:** Execute a full RISC-V compliance suite or compiler-generated code to verify all instructions (integer, floating-point, CSR, exceptions). For vector units, include stream/vector workloads (e.g. vector add/multiply loops).

4. **Corner Cases:** Memory stress (random accesses, concurrent loads/stores), branch/divergence, interrupt handling, atomic instructions, timer and I/O interrupts.
5. **Performance Benchmarks:** Run benchmarks like CoreMark or Dhrystone (adapted for RISC-V) to measure IPC and frequency. Also test any DSP kernels or AI kernels the vector unit targets.
6. **Functional Checks:** Compare FPGA outputs against a golden model (QEMU/Rocket-Chip reference or software simulator) for cycle-accuracy on key workloads. Use waveform dumps or monitor variables (via on-chip debug) to ensure coherence events behave correctly.
7. **Debug Infrastructure:** Implement simple debug: JTAG or a UART console to print status. Use internal logic analyzers (FPGA IP) to capture failures.

This validation provides coverage and confidence that the synthesizable RTL matches the intended architecture. It also reveals any missing functionality or mis-timings before expensive silicon.

IP Licensing and Third-Party Dependencies

- **Inventory All IP:** List every third-party or foundry-provided module. Common examples: PHY (Ethernet, DDR, SerDes), PLL/DLLs, analog (bandgap, regulators), EEPROM or flash, cryptographic blocks, RISC-V core (if imported), and any macros (UART, GPIO).
- **Check Licenses:** For each IP, note license terms. Open-source IP (MIT/BSD/GPL) may have conditions; proprietary IP requires NDA and licensing fee. For instance, **X-FAB** explicitly requires an IP license agreement for its blocks【31†L1082-L1086】.
- **Verify IP Pin Compatibility:** Ensure any IP block matches target PDK I/O standards (voltage levels, pad pitch).
- **Document Version/Date:** Record source and version of IP (e.g. "RISC-V core v2.1 by OpenHW" or "DDR PHY version 3.4 from Cadence") and any toolchain used to generate it.
- **Plan for Integration:** Some IP may come as encrypted LEF/DEF (e.g. ARM macros). Ensure the design team has decryption keys/PIN locks and the EDA tools can handle them.

A concise **IP Checklist** might include items like:

- Verify RISC-V core license (Apache 2.0, no liability).
- Obtain DDR4/AXI memory controller (license fee).
- License PCIe/Ethernet PHYs (if needed).
- Confirm PLL or HV IO licenses (foundry may bundle these).
- Check any patents (e.g. algorithms in DSP) for compliance.

Prioritized Work Programme and Milestones

Below is a high-level roadmap to achieve a synthesis-ready RTL release (estimates assume an experienced ASIC team). Actual costs depend on team rates and tools chosen.

1. **RTL Audit and Conversion (4–6 weeks)**

2. **Milestone:** All `.sf` sources converted to SV; no syntax errors in synthesis.

3. **Tasks:** Translate high-level files, remove unsynthesizable constructs, refactor FSMs into RTL form. Run lint and fix issues.

4. **Cost:** ~2–3 engineers × 1 month ≈ £30k per engineer, total ~~£80k.

5. **Memory and IP Integration (4 weeks)**

6. **Milestone:** All cache/memory blocks instantiated with real SRAM macros or compiler output; DDR/AXI interface validated.

7. **Tasks:** Source or generate SRAM macros for each cache/buffer; insert them into RTL. Integrate vendor IP (e.g. DDR PHY) and adapt interfaces. Secure any new IP licenses.

8. **Cost:** Memory compiler licenses or custom macro development (~£20k), plus engineering (~£50k).

9. **Verification Closure (6–8 weeks)**

10. **Milestone:** Complete self-checking testbenches and coverage reports.

11. **Tasks:** Build directed tests (ISA, vector ops, interconnect), set up regression framework, fill coverage holes, simulate with synthesis netlist (gate-level).

12. **Cost:** ~2 engineers × 2 months = ~~£120k.

13. **FPGA Prototyping (6 weeks)**

14. **Milestone:** SilicaFlux-V (reduced core) runs on FPGA at functional speed.

15. **Tasks:** Synthesize for FPGA, create bitstream, debug on hardware. Use of FPGA boards (~£5k for boards).

16. **Cost:** FPGA tools (Vivado license ~$10k), boards + engineering (~£50k).

17. **Physical Design Prep (8 weeks)**

18. **Milestone:** Floorplan complete, all physical constraints set, run initial P&R (timing at target f_{max} annotated).

19. **Tasks:** Choose foundry & PDK; import PDK libraries; define voltage domains; place SRAM macros; power grid design. Generate SDC constraints.

20. **Cost:** PDK license (~£10k via Europractice for MPW), EDA license or foundry bundle (£30k). 2 engineers × 2 months (~£100k).

21. **Tapeout and Fabrication**

22. **Milestone:** GDSII submitted, wafers returned (engineering samples).

23. **Cost:** Foundry masks and wafers (see below), assembly & test.

Budget Estimates:

- **EDA Tools:** Commercial suite (synthesis, P&R, STA, DRC/LVS) ≈ £50k–£100k per seat/year. Possible open-source substitution (Yosys, OpenROAD) for early phases.
- **Foundry (MPW 28 nm):** Using Europractice pricing【29†L7-L10】, GF 28 SLPe masks at **€12,430/mm²** standard. For a conservative 60 mm² die, this is ~€746k for masks alone (discounted ~€678k).
- **Wafers & Packaging:** Europractice "fabrication" fee (~€20k) covers 50 dies (unpackaged). Bump packaging (~€150/die) for 50 pcs ≈ €7.5k.
- **Assembly & Test:** Probe/test of wafers (~€15k), plus automated test development (~£20k).
- **Contingency:** Additional spin (~€500k) should be budgeted for unexpected respins or IP licensing.

A sample **Foundry RFQ Template** line might read: "**Process:** GF 28SLPe (6L SRAM) MPW – Target f_{max} >1 GHz. **Mask cost:** as per Europractice 2026 pricing (~€12,430/mm²【29†L7-L10】).

Minimum area: 10 mm^2 (Europractice rule). **Services:** 50 engineering samples (flip-chip BGA package), probe test, one wirebonded package if needed. NDA/IP support included."

Checklists and Next Steps

- **RTL Sanity:** Verify coding guidelines (no inferred latches, one clock per `always_ff`, no wildcard sensitivity, etc). Run a synthesizer and fix errors iteratively.
- **DFT Plan:** Insert scan chains and optionally BIST for memories. Decide on JTAG or IEEE 1500 for test access.
- **Clocking Scheme:** Finalize clock-tree plan (one global clock or multiple? Gating? Meshing?). Ensure each domain has a reset synchronizer.
- **DFM & Reliability:** Include simulation for IR drop and electromigration (after P&R) to ensure power rails are robust.
- **Verification Signoff:** Have at least 90% code coverage and all assertions passing in formal checks.
- **Documentation:** Update design docs (e.g. the Silica Flux PDF) with any RTL changes. Maintain versioned release notes for RTL.

In conclusion, while the path to a fabricated SilicaFlux ASIC is clear (and follow the foundry routes via Europractice/IC-Link【11†L39-L47】【26†L150-L159】), significant RTL and design work remains. The first priorities are to achieve a fully synthesizable RTL baseline and a working FPGA prototype. Only after passing that milestone should one commit to a 28 nm MPW. With disciplined project management, we estimate **6–9 months** of engineering effort (plus queuing for foundry) to reach a first silicon. Building on the steps above and citing sources like Europractice pricing【29†L7-L10】 and AMD guidelines 【48†L513-L521】 helps ensure that decision-makers have a factual, data-driven roadmap to go from "pure silica" design files to working silicon.

Sources: Verified foundry schedules and pricing【29†L7-L10】【15†L240-L244】, vendor guidelines on synthesizability and resets【19†L244-L252】【48†L513-L521】, open-source memory compiler description【50†L241-L244】, and Europractice IP notes【31†L1082-L1086】 were used to inform this audit and recommendations. (Further details can be provided from the SilicaFlux code/README and foundry PDK docs as needed.)
